

Yuyou Zhang

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RESEARCH INTERESTS

- **Vision-Language Models (VLMs)** [1, 2]: Spatial and temporal reasoning, perspective taking, 3D world understanding, and diagnostic benchmark design for multimodal models.
- **Large Language Models (LLMs)** [3, 4]: Safety, alignment, and responsible AI, including reasoning-based safety mechanisms.
- **Embodied AI & Reinforcement Learning** [5, 6, 7, 8, 9, 10, 11]: Learning and control for embodied agents, with a focus on robust decision-making, adaptability, and safety in dynamic real-world environments.

EDUCATION

Carnegie Mellon University

Ph.D. in Mechanical Engineering, Robotics focus, Safe AI Lab

Pittsburgh, USA

Aug. 2022 - May 2027 (expected)

Shanghai Jiao Tong University

B.Eng., Automation, School of Electronic Information Electrical Engineering

Shanghai, China

B.Eng., ZhiYuan Honors Program, Zhiyuan College

Sept. 2018 - June 2022

SELECTED PUBLICATIONS

- [1] **Y. Zhang**, R. Corcodel, C. Hori, A. Cherian, and D. Zhao, “Spinbench: Perspective and rotation as a lens on spatial reasoning in vlms,” *Best Paper Runner Up NeurIPS SPACE in Vision, Language, and Embodied AI Workshop*, 2025.
- [2] M. Li, W. Ding, H. Lin, Y. Lyu, Y. Yao, **Y. Zhang**, and D. Zhao, “Crashagent: Crash scenario generation via multi-modal reasoning,” *arXiv*, 2025.
- [3] **Y. Zhang**, M. Li, W. Han, Y. Yao, Z. Cen, and D. Zhao, “Safety is not only about refusal: Reasoning-enhanced fine-tuning for interpretable llm safety,” *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
- [4] Y. Yao, Z. Cen, M. Li, W. Han, **Y. Zhang**, E. Liu, Z. Liu, C. Gan, and D. Zhao, “Your language model may think too rigidly: Achieving reasoning consistency with symmetry-enhanced training,” *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
- [5] **Y. Zhang**, Y. Yao, S. Liu, Y. Niu, C. Lin, Y. Yang, W. Yu, T. Zhang, J. Tan, and D. Zhao, “Quietpaw: Learning quadrupedal locomotion with versatile noise preference alignment,” *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [6] **Y. Zhang**, R. Corcodel, and D. Zhao, “Bipedalism for Quadrupedal Robots: Versatile Loco-Manipulation through Risk-Adaptive Reinforcement Learning,” in *IEEE-RAS International Conference on Humanoid Robots*, Sept. 2025.
- [7] **Y. Zhang**, Y. Niu, X. Liu, and D. Zhao, “Composer: Scalable and robust modular policies for snake robots,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10800–10806, IEEE, 2024.
- [8] C. Wang, **Y. Zhang**, X. Zhang, Z. Wu, X. Zhu, S. Jin, T. Tang, and M. Tomizuka, “Offline-online learning of deformation model for cable manipulation with graph neural networks,” *IEEE Robotics and Automation Letters (RAL)*, vol. 7, no. 2, pp. 5544–5551, 2022.
- [9] F. Xu, **Y. Zhang**, J. Sun, and H. Wang, “Adaptive visual servoing shape control of a soft robot manipulator using bézier curve features,” *IEEE/ASME Transactions on Mechatronics*, 2022.
- [10] L. Han, **Y. Zhang**, and H. Wang, “Hybrid adaptive vision-force control under the bottleneck constraint,” *IEEE Transactions on Control Systems Technology*, 2022.

- [11] L. Han, **Y. Zhang**, and H. Wang, “Vision-based contact point selection for the fully non-fixed contact manipulation of deformable objects,” *IEEE Robotics and Automation Letters (RAL)*, vol. 7, no. 2, pp. 4368–4375, 2022.

RESEARCH EXPERIENCE

Safe AI Lab

Ph.D Candidate

Carnegie Mellon University

Aug 2022 - present

SpinBench: Perspective and Rotation as a Lens on Spatial Reasoning in VLMs [1]

- SpinBench is a cognitively grounded diagnostic benchmark for evaluating spatial reasoning in vision language models (VLMs).
- SpinBench targets translation, rotation, object relative pose, and viewpoint change.
- Progressively structured: single-object tasks scaffold toward multi-object perspective-taking tasks.

Safety is Not Only About Refusal: Reasoning-Enhanced Fine-tuning for Interpretable LLM Safety [3]

- Proposed a novel framework that trains models to engage in explicit safe reasoning before response.
- Reasoning is not only a key component of model capability but also powerful for safety alignment.

QuietPaw: Learning Quadrupedal Locomotion with Versatile Noise Preference Alignment [5]

- Proposed a conditional RL framework for noise-constrained locomotion.
- A decomposed value estimation approach that improves Pareto-optimality.

Bipedalism for Quadrupedal Robots: versatile loco-manipulation through risk-adaptive RL [6]

- Proposed a risk-adaptive RL framework for the robust bipedal locomotion of quadrupedal robots
- Introduced an uncertainty metric based on the return distribution for risk awareness
- Robust loco-manipulations with cart pushing, contact-aware obstacle probing, payload carrying

COMPOSER: scalable and robust modular policies for snake robots [7]

- Proposed a modular policy with a self-attention mechanism to enhance cooperative behaviors
- Demonstrated superior efficiency, robustness against corruption, and zero-shot generalizability.

Mechanical Systems Control Lab

Undergraduate Researcher

University of California, Berkeley

June 2021 - Nov. 2021

Offline-online learning of deformation model for cable manipulation with graph neural networks [8]

- Combined offline graph neural network with online residual model to approximate cable dynamics
- Improved model training efficiency and generalizability, narrowed sim-to-real gap.

Intelligent Robotics and Machine Vision Lab

Undergraduate Researcher

Shanghai Jiao Tong University

Sept. 2020 - June 2022

Safe Vision-based contact selection for the non-fixed contact manipulation of deformable objects[11]

- Proposed a stabilizing and optimization strategy to select initial manipulation contact points
- Designed an uncalibrated visual servo controller to validate the contact optimization strategy

Hybrid vision-force control for robotics manipulation in confined space[10]

- Developed an adaptive method to estimate Jacobian matrixes in force space and image space
- Designed a hybrid vision-force controller for manipulation with a desired contact force

Adaptive visual servoing shape control of a soft robot manipulator using Bézier curve features[9]

- Proposed an adaptive Bézier curve update algorithm to represent soft continuum robots
- Designed an uncalibrated visual servo controller for shape regulation of the soft continuum robot

WORKING EXPERIENCE

Mitsubishi Electric Research Laboratories

Research Intern

May. 2025 - Aug. 2025

- Proposed **SpinBench**, a diagnostic benchmark for spatial reasoning in VLMs.
- Conducted large-scale evaluation of **43+ VLMs** across spatial transformation tasks.
- **Fine-tuned InternVL** on curated data, improving spatial reasoning performance.

Mitsubishi Electric Research Laboratories

Research Intern

May. 2024 - Aug. 2024

- Developed distributional risk-aware policy for quadrupedal robot locomotion
- Proposed bipedal locomotion for quadrupedal robots to enable flexible loco-manipulation.

Flexiv Robotics Inc.*Robotic Vision Intern**Dec. 2021 - Mar. 2022*

- 3D reconstruction and point cloud registration
- Point cloud feature extraction and visual fixture generation for interactive teleoperation

Shanghai Inspection and Testing Institute of Instruments and Automation Systems Co. Ltd.*Engineer Intern**June. 2021 - Aug. 2021*

TECHNICAL SKILLS

ML: Vision–Language Models, Large Language Models, Multimodal Learning, Model Evaluation, Fine-tuning, Benchmark Design, Reinforcement Learning, Distributional RL, Imitation Learning

Frameworks: PyTorch, HuggingFace

Robotics: Locomotion, Loco-manipulation, Visual Servoing

AWARDS

- CMU Graduate Student Assembly Travel Grants for ICRA 2024
- Carnegie Institute of Technology Dean's Fellowship, 2022
- Shanghai Outstanding Graduate, Top 2.5%
- Zhiyuan Honors Degree, Top 1.5%
- Zhiyuan Honors Scholarship, 2018, 2019, 2020, 2021, Top 5%
- Undergraduate Excellence Scholarship 2018, 2019, 2020, 2021
- The Mathematical Contest in Modeling: Meritorious Winner, Top 3%